

Introduction to Topology II: Homework 6

Due on February 18, 2026 at 23:59

Nick Proudfoot 13:00

Hashem A. Damrah

UO ID: 952102243

Problem 1. Prove Lemma 8.8 from the book (also stated in class), which says that the map χ is a chain map.

Solution. We prove that the map χ is a chain map, i.e.

$$\partial \circ \chi = \chi \circ \partial.$$

Let $\sigma : \Delta^n \rightarrow X$ be a singular n -simplex. By definition,

$$\chi(\sigma) = \sum_{i=0}^n (-1)^i \sigma|_{[v_0, \dots, \widehat{v_i}, \dots, v_n]},$$

where the restriction is taken to the i th face of Δ^n .

Applying the boundary operator, we obtain

$$\partial(\chi(\sigma)) = \sum_{i=0}^n (-1)^i \partial(\sigma|_{[v_0, \dots, \widehat{v_i}, \dots, v_n]}).$$

Each restricted map $\sigma|_{[v_0, \dots, \widehat{v_i}, \dots, v_n]}$ is an $(n-1)$ -simplex, so its boundary is

$$\partial(\sigma|_{[v_0, \dots, \widehat{v_i}, \dots, v_n]}) = \sum_{j=0}^{n-1} (-1)^j \sigma|_{[v_0, \dots, \widehat{v_i}, \dots, \widehat{v_k}, \dots, v_n]},$$

where k denotes the original vertex omitted after removing the i th vertex.

Hence,

$$\partial(\chi(\sigma)) = \sum_{i < k} (-1)^{i+k-1} \sigma|_{[v_0, \dots, \widehat{v_i}, \dots, \widehat{v_k}, \dots, v_n]} + \sum_{k < i} (-1)^{i+k} \sigma|_{[v_0, \dots, \widehat{v_k}, \dots, \widehat{v_i}, \dots, v_n]}.$$

Each codimension-2 face appears exactly twice with opposite signs, so the terms cancel in pairs. Therefore, $\partial(\chi(\sigma)) = 0$.

On the other hand,

$$\partial\sigma = \sum_{i=0}^n (-1)^i \sigma|_{[v_0, \dots, \widehat{v_i}, \dots, v_n]}.$$

Applying χ ,

$$\chi(\partial\sigma) = \sum_{i=0}^n (-1)^i \chi(\sigma|_{[v_0, \dots, \widehat{v_i}, \dots, v_n]}).$$

Expanding this produces the same collection of codimension-2 faces with the same coefficients as above, and hence the same cancellations occur. Therefore, $\chi(\partial\sigma) = 0$. Thus,

$$\partial(\chi(\sigma)) = \chi(\partial\sigma),$$

and since both ∂ and χ are linear, this equality holds for all chains. Hence χ is a chain map. $\square$

Problem 2. Suppose that $s : |K^m| \rightarrow |L|$ is a simplicial approximation of $f : |K| \rightarrow |L|$, $n > m$, and $\theta : |K^n| \rightarrow |K^m|$ is a standard simplicial map (page 188). Show that $s \circ \theta : |K^n| \rightarrow |L|$ is also a simplicial approximation of f .

Solution. Recall that $s : |K^m| \rightarrow |L|$ is a simplicial approximation to $f : |K| \rightarrow |L|$ if for every vertex v of K^m ,

$$f(\text{St}(v)) \subset \text{St}(s(v)),$$

where $\text{St}(v)$ denotes the open star of v .

Let $\theta : |K^n| \rightarrow |K^m|$ be the standard simplicial map with $n > m$. We must show that the composition

$$s \circ \theta : |K^n| \rightarrow |L|,$$

is also a simplicial approximation to f . That is, for each vertex w of K^n , we must verify

$$f(\text{St}(w)) \subset \text{St}((s \circ \theta)(w)).$$

Let w be a vertex of K^n . Since θ is simplicial, it maps vertices of K^n to vertices of K^m , and it carries simplices linearly onto simplices. In particular, there exists a vertex $v = \theta(w)$ of K^m such that

$$\theta(\text{St}(w)) \subset \text{St}(v).$$

Now let $x \in \text{St}(w)$. Then $\theta(x) \in \text{St}(v)$. Since s is a simplicial approximation to f , we have

$$f(\text{St}(v)) \subset \text{St}(s(v)).$$

Because $x \in \text{St}(w)$ implies $\theta(x) \in \text{St}(v)$, it follows that

$$f(x) \in \text{St}(s(v)).$$

But $v = \theta(w)$, so $s(v) = s(\theta(w)) = (s \circ \theta)(w)$. Therefore,

$$f(x) \in \text{St}((s \circ \theta)(w)).$$

Since this holds for all $x \in \text{St}(w)$, we conclude

$$f(\text{St}(w)) \subset \text{St}((s \circ \theta)(w)).$$

Hence $s \circ \theta$ is a simplicial approximation to f . □

Problem 3. Suppose that $s, t : |K| \rightarrow |L|$ are simplicial maps, and let $s_i, t_i : C_i(K) \rightarrow C_i(L)$ denote the induced maps on oriented i -simplices. Suppose also that, for each i , we have a linear map $d_i : C_i(K) \rightarrow C_{i+1}(L)$ with the property that we have the following equality of linear maps from $C_i(K)$ to $C_i(L)$:

$$d_{i-1} \circ \partial_i^K + \partial_{i+1}^L \circ d_i = t_i - s_i,$$

Such a collection of maps d_i is called a *chain homotopy* relating the simplicial maps s and t . Show that we have $t_* = s_* : H_i(K) \rightarrow H_i(L)$.

Solution. Let $s_i, t_i : C_i(K) \rightarrow C_i(L)$ be the chain maps induced by the simplicial maps s and t , and suppose there exist linear maps

$$d_i : C_i(K) \rightarrow C_{i+1}(L),$$

such that

$$d_{i-1} \circ \partial_i^K + \partial_{i+1}^L \circ d_i = t_i - s_i.$$

We must show that the induced maps on homology satisfy

$$t_* = s_* : H_i(K) \rightarrow H_i(L).$$

Let $[z] \in H_i(K)$, where $z \in C_i(K)$ is a cycle, so

$$\partial_i^K z = 0.$$

Apply the chain homotopy identity to z :

$$(t_i - s_i)(z) = d_{i-1}(\partial_i^K z) + \partial_{i+1}^L(d_i z).$$

Since z is a cycle, the first term vanishes:

$$d_{i-1}(\partial_i^K z) = 0.$$

Hence,

$$t_i(z) - s_i(z) = \partial_{i+1}^L(d_i z).$$

This shows that $t_i(z) - s_i(z)$ is a boundary in $C_i(L)$. Therefore $t_i(z)$ and $s_i(z)$ represent the same homology class:

$$[t_i(z)] = [s_i(z)] \in H_i(L).$$

By definition of the induced maps on homology,

$$t_*([z]) = [t_i(z)], \quad s_*([z]) = [s_i(z)],$$

and thus

$$t_*([z]) = s_*([z]).$$

Since this holds for every homology class $[z] \in H_i(K)$, we conclude that

$$t_* = s_* : H_i(K) \rightarrow H_i(L). \quad \square$$

Problem 4. Recall that two simplicial maps $s, t : |K| \rightarrow |L|$ are *close* if, for every simplex $A \in K$, there exists a simplex $B \in L$ such that $s(A)$ and $t(A)$ are both faces of B . The smallest such simplex B is called the *carrier* of A . The goal of this problem is to show that, if s and t are close, then they are related by a chain homotopy, and therefore they induce the same map on homology. Assume that s and t are close.

(i) Define $d_0 : C_0(K) \rightarrow C_1(L)$ by putting

$$d_0(v) = \begin{cases} 0 & \text{if } s(v) = t(v) \\ (s(v), t(v)) & \text{otherwise.} \end{cases}$$

Show that $\partial_1^L \circ d_0 = t_0 - s_0 : C_0(K) \rightarrow C_0(L)$.

(ii) Suppose that $j > 0$, and that, for all $i < j$, we have defined homomorphisms $d_i : C_i(K) \rightarrow C_{i+1}(L)$ such that

$$d_{i-1} \circ \partial_1^K + \partial_{i+1}^L \circ d_i = t_i - s_i,$$

we have defined homomorphisms $\sigma, d_i(\sigma)$ is always a chain in the carrier of σ . Show that, for any oriented j -simplex σ of K , we have

$$\partial_j^L(t_j(\sigma) - s_j(\sigma) - d_{j-1} \circ \partial_j^K(\sigma)) = 0.$$

(iii) Let $M \subset L$ be the subcomplex of L consisting of the carrier of σ and all of its faces. Show that there exists an element $c \in C_{j+1}(M) \subset C_{j+1}(L)$ such that

$$t_j(\sigma) - s_j(\sigma) - d_{j-1} \circ \partial_j^K(\sigma) = \partial_{j+1}^L(c).$$

(iv) Finish the proof that the simplicial maps s and t are related by a chain homotopy.

Solution to (i). Let v be a vertex of K . If $s(v) = t(v)$, then by definition $d_0(v) = 0$, and hence

$$\partial_1^L(d_0(v)) = 0 = t_0(v) - s_0(v).$$

If $s(v) \neq t(v)$, then

$$d_0(v) = (s(v), t(v)),$$

an oriented 1-simplex of L . Applying the boundary operator,

$$\partial_1^L(s(v), t(v)) = t(v) - s(v).$$

Thus in this case as well,

$$\partial_1^L(d_0(v)) = t_0(v) - s_0(v).$$

Since both sides are linear in v , it follows that

$$\partial_1^L \circ d_0 = t_0 - s_0 : C_0(K) \rightarrow C_0(L). \quad \square$$

Solution to (ii). Let σ be an oriented j -simplex of K , where $j > 0$. Consider

$$z = t_j(\sigma) - s_j(\sigma) - d_{j-1}(\partial_j^K(\sigma)).$$

We compute its boundary. Since s_* and t_* are simplicial maps, they are chain maps, so

$$\partial_j^L \circ t_j = t_{j-1} \circ \partial_j^K \quad \text{and} \quad \partial_j^L \circ s_j = s_{j-1} \circ \partial_j^K.$$

Therefore,

$$\begin{aligned} \partial_j^L z &= \partial_j^L t_j(\sigma) - \partial_j^L s_j(\sigma) - \partial_j^L d_{j-1}(\partial_j^K(\sigma)) \\ &= t_{j-1}(\partial_j^K(\sigma)) - s_{j-1}(\partial_j^K(\sigma)) - \partial_j^L d_{j-1}(\partial_j^K(\sigma)). \end{aligned}$$

By the inductive hypothesis for $i = j - 1$,

$$d_{j-2} \circ \partial_{j-1}^K + \partial_j^L \circ d_{j-1} = t_{j-1} - s_{j-1}.$$

Applying this to $\partial_j^K(\sigma)$ gives

$$t_{j-1}(\partial_j^K(\sigma)) - s_{j-1}(\partial_j^K(\sigma)) = d_{j-2}(\partial_{j-1}^K \partial_j^K(\sigma)) + \partial_j^L d_{j-1}(\partial_j^K(\sigma)).$$

Since $\partial_{j-1}^K \circ \partial_j^K = 0$, this reduces to

$$t_{j-1}(\partial_j^K(\sigma)) - s_{j-1}(\partial_j^K(\sigma)) = \partial_j^L d_{j-1}(\partial_j^K(\sigma)).$$

Substituting into the expression for $\partial_j^L z$, we obtain

$$\partial_j^L z = 0.$$

Hence

$$\partial_j^L (t_j(\sigma) - s_j(\sigma) - d_{j-1}(\partial_j^K(\sigma))) = 0. \quad \square$$

Solution to (iii). Let B be the carrier of σ , and let $M \subset L$ be the subcomplex consisting of B and all of its faces. Since s and t are close, both $s(\sigma)$ and $t(\sigma)$ are faces of B . By the inductive assumption, $d_{j-1}(\tau)$ lies in the carrier of each face τ of σ , and hence also lies in M . Therefore

$$z = t_j(\sigma) - s_j(\sigma) - d_{j-1}(\partial_j^K(\sigma)),$$

is a chain in $C_j(M)$.

By part (ii), $\partial_j^L z = 0$, so z is a cycle in $C_j(M)$. Since M is a simplex (together with its faces), it is contractible, and hence

$$H_j(M) = 0 \quad \text{for } j > 0.$$

Therefore every j -cycle in M is a boundary. Thus there exists

$$c \in C_{j+1}(M) \subset C_{j+1}(L),$$

such that

$$z = \partial_{j+1}^L(c).$$

That is,

$$t_j(\sigma) - s_j(\sigma) - d_{j-1} \circ \partial_j^K(\sigma) = \partial_{j+1}^L(c). \quad \square$$

Solution to (iv). For each oriented j -simplex σ of K , define

$$d_j(\sigma) = c,$$

where c is the chain provided by part (iii). Extend d_j linearly to all of $C_j(K)$. By construction, $d_j(\sigma)$ lies in the carrier of σ .

The defining property from part (iii) then gives

$$t_j(\sigma) - s_j(\sigma) = d_{j-1}(\partial_j^K(\sigma)) + \partial_{j+1}^L(d_j(\sigma)).$$

Since both sides are linear, this equality holds for all chains in $C_j(K)$. Hence

$$d_{j-1} \circ \partial_j^K + \partial_{j+1}^L \circ d_j = t_j - s_j.$$

By induction, we obtain maps d_i for all $i \geq 0$ satisfying

$$d_{i-1} \circ \partial_i^K + \partial_{i+1}^L \circ d_i = t_i - s_i.$$

Thus $\{d_i\}$ is a chain homotopy relating the simplicial maps s and t . Therefore s and t induce the same maps on homology. $\square$